Aman Tayal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.55
Disha Delphi Public School	Class XII (CBSE)	2024	92.40%
Delhi Public School	Class X (CBSE)	2022	95.83%

SKILLS

Programming Languages: C, C++, Python and SQL

Data Analysis & ML: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

GenAI & RAG: Sentence Transformers, PostgreSQL, pgvector, Google Gemini

Robotics: ROS2, Gazebo, Navigation2, URDF/Xacro

Tools: Git, GitHub, Linux (Ubuntu), Jupyter Notebook, VS Code, MySQL

RELEVANT COURSES

Introduction to Programming (ID1063)

Digital Fabrication(ID1054)

Corporate Finance

PROJECTS

Document RAG Assistant | Python, PostgreSQL, pgvector, Sentence Transformers, Gemini

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline for querying technical PDF documents using semantic search.
- Implemented PDF extraction, overlapping text chunking, **Sentence Transformer embeddings**, and vector storage/retrieval using **PostgreSQL + pgvector**.
- Integrated **Google Gemini** to generate grounded answers from retrieved document context with **source page citations**.

Credit Card Default Prediction & Model Analysis | Python, pandas, NumPy, scikit-learn, Matplotlib, Seaborn

- Analyzed 30,000 credit-card customer records, performing data cleaning, exploratory analysis, correlation analysis, and default-rate analysis.
- Built and evaluated Logistic Regression, Decision Tree, and Random Forest classification models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- Compared model trade-offs, with Random Forest achieving 81.33% accuracy while Decision Tree achieved 59.91% recall for identifying defaulters.
- Investigated feature importance and repayment behavior, identifying recent repayment status as a strong indicator of default risk.

ROS2 Autonomous Maze Robot | ROS2 Humble, Gazebo, SLAM Toolbox, Navigation2, URDF/Xacro

- Designed and simulated a custom differential-drive mobile robot using ROS2 and Gazebo.
- Integrated LiDAR and implemented SLAM-based occupancy grid mapping using SLAM Toolbox.
- Developed a complete simulation pipeline including robot modeling, teleoperation, sensor integration, and map generation.
- Integrated the Navigation2 stack for autonomous navigation (currently under development).

POSITIONS OF RESPONSIBILITY

Mechatronics Core in Tinkerer's Lab